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Research Article

Influence of Casting Route on the Microstructure and Mechanical Behavior of A360 Aluminum Alloy: A Process Structure Property Correlation

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Abstract: This study comprehensively examines the effect of casting route sand versus permanent mold on the microstructural evolution and mechanical performance of A360 aluminum alloy produced under identical alloy composition. The objective was to elucidate how differences in cooling rate and solidification kinetics affect dendritic morphology, eutectic silicon distribution, and the resulting property–structure correlation. The alloy was melted at 730 ± 5 °C under an argon protected atmosphere and cast using silica-bentonite sand and preheated steel molds (200 °C) under gravity conditions to ensure reproducibility. Optical microscopy revealed that the permanent mold route generated a highly refined α-Al dendritic structure with reduced secondary dendrite arm spacing (SDAS) and a uniformly dispersed eutectic Si network, whereas the sand cast alloy exhibited coarse dendrites and irregular Si clusters. These microstructural distinctions led to clear performance differences: the permanent mold alloy achieved approximately 18 % higher Brinell hardness and 10–15 % higher tensile strength while maintaining comparable elongation. Furthermore, its Charpy impact energy increased from 4.81 J to 5.07 J, reflecting superior energy absorption and fracture toughness due to finer and more homogeneous solidification morphology. The findings emphasize that controlling the casting process specifically through enhanced cooling rate can significantly strengthen A360 alloy without any chemical modification or heat treatment. This study provides fundamental insight into the process–structure–property relationship and offers a practical reference for sustainable design and manufacturing of high-integrity aluminum components for automotive and aerospace applications.

Keywords: *A360 aluminum alloy, hardness, impact energy, microstructure, permanent mold casting, sand mold casting, tensile strength.*

Kalıp Tipinin A360 Alüminyum Alaşımının Mikroyapı ve Mekanik Davranışı Üzerindeki Etkisi: Süreç Yapı Özellik İlişkisi

Öz. Bu çalışma, kimyasal bileşimi aynı tutulan A360 alüminyum alaşımında, farklı döküm yöntemlerinin kum kalıba ve kokil kalıba döküm mikroyapısal evrim ve mekanik özellikler üzerindeki etkisini kapsamlı biçimde incelemiştir. Çalışmanın temel amacı, soğuma hızı ve katılaşma kinetiğindeki farklılıkların dendrit morfolojisi, ötektik silisyum dağılımı ve buna bağlı olarak oluşan yapı–özellik ilişkisi üzerindeki etkilerini ortaya koymaktır. Alaşım, 730 ± 5 °C'de argon atmosferi altında ergitilmiş ve karşılaştırılabilir deney koşullarını sağlamak amacıyla silika–bentonit esaslı kum kalıplar ile 200 °C'de ön ısıtılmış çelik kokil kalıplara yerçekimi yöntemiyle dökülmüştür. Optik mikroskop analizleri, kokil kalıba döküm yöntemiyle elde edilen numunelerde daha ince α-Al dendritleri, azalmış ikincil dendrit kol aralığı (SDAS) ve daha homojen dağılmış ötektik Si ağı yapısının oluştuğunu; buna karşın kum kalıba döküm

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numunelerinde daha kaba dendritik yapı ve düzensiz Si kümelenmeleri gözlendiğini ortaya koymuştur. Bu mikroyapısal farklılıklar, mekanik özelliklerde belirgin farklılıklar yaratmıştır: kokil kalıba döküm yöntemiyle üretilen alaşım, Brinell sertliğinde yaklaşık %18 ve çekme dayanımında %10–15 oranında artış göstermiştir. Ayrıca darbe enerjisi 4,81 J'den 5,07 J'ye yükselmiş olup, bu artış daha ince ve homojen bir katılaşma yapısına bağlı olarak gelişen daha yüksek enerji soğurma ve kırılma tokluğunu yansıtmaktadır. Elde edilen sonuçlar, döküm sürecinin kontrol edilmesinin özellikle soğuma hızının artırılmasının herhangi bir kimyasal modifikasyon veya ısıl işlem uygulanmadan A360 alaşımının mukavemetini önemli ölçüde artırabileceğini göstermektedir. Bu çalışma, süreç–yapı–özellik ilişkisine dair temel bir bilimsel bakış açısı sunmakta ve yüksek bütünlüğe sahip alüminyum bileşenlerin otomotiv ve havacılık uygulamaları için sürdürülebilir üretim tasarımlarına pratik bir referans oluşturmaktadır.

Anahtar kelimeler: *A360 alüminyum alaşımı, çekme dayanımı, darbe enerjisi, mikroyapı, kokil kalıp döküm, kum kalıp döküm, sertlik.*

1. Introduction

Aluminum casting alloys particularly the Al–Si series including hypoeutectic Al–Si series (A356, A360, A357) constitute the cornerstone of many modern engineering applications owing to their unique combination of low density, high specific strength, and excellent thermal stability [1-4]. Their outstanding castability and fluidity enable the large-scale production of complex automotive and aerospace components such as engine blocks, cylinder heads, and transmission housings [3,5,6].

Among this family, the A360 aluminum alloy is of special industrial significance due to its relatively high silicon and magnesium content, which provides a superior balance between strength, ductility, and resistance to thermal fatigue [7-10]. Consequently, A360 plays a strategic role in structural applications that demand dimensional stability and long-term reliability under cyclic thermal and mechanical loading conditions [7,8,9,11].

The overall performance and service life of metallic materials are determined by the intrinsic relationship between process, structure, and properties [12-14]. Even when the alloy chemistry remains constant, the choice of manufacturing process can dramatically alter solidification kinetics, which in turn governs microstructural evolution and mechanical response [15-17]. In casting processes, the thermal history dictates grain refinement, secondary dendrite arm spacing (SDAS), eutectic silicon morphology, and porosity distribution all of which are decisive factors influencing the alloy's strength, ductility, and toughness [17-19].

From this perspective, Al–Si–Mg alloys have been extensively studied and employed across the automotive, marine, and aerospace sectors due to their advantageous combination of castability and performance [1,2,20]. The rate of heat extraction during solidification, which is inherently governed by the casting route, has a profound influence on microstructural development [21,22]. Sand molds, characterized by low thermal conductivity, typically produce coarser α-Al dendrites and irregular eutectic silicon morphologies. In contrast, permanent molds enable faster heat dissipation, promoting refined dendritic networks, smaller SDAS, and a more homogeneous distribution of eutectic phases [15,17,21,22].

Numerous studies have demonstrated that higher solidification cooling rates enhance the mechanical performance of Al–Si casting alloys by refining the secondary dendrite arm spacing (SDAS) and modifying the eutectic silicon morphology [21–26]. In both high-pressure die casting (HPDC) and gravity casting contexts, an increase in cooling rate has been reported to produce significant improvements in strength-related properties [21,22,25,26]. These improvements are primarily attributed to the refinement and spheroidization of the eutectic Si phase, which alleviate local stress concentrations and delay crack initiation [23,27]. Furthermore, the refinement of the overall microstructure and the controlled distribution of cathodic secondary phases such as Si and Fe-containing intermetallic compounds reduce micro-galvanic coupling with the α-Al matrix. As a result, the tendency toward localized corrosion is diminished, and both mechanical integrity and corrosion resistance are simultaneously enhanced [20,24,28,30].

Despite these advances, the majority of previous research has focused primarily on chemical modification such as Sr, Na, or Ti additions to refine eutectic silicon and improve mechanical performance [8,13,31,32]. However, the intrinsic effect of the casting route itself**,** independent of alloying additions, has received comparatively less attention. Nevertheless, despite the extensive focus on chemical modification strategies, comparative insights into process-controlled phenomena remain underexplored. Comparative studies addressing the microstructural and mechanical evolution of A360 alloy under different casting conditions remain scarce [7,8,9,33]. This limitation makes it difficult for engineers and foundries to access reliable quantitative data when selecting the most appropriate casting process for specific component requirements [12,16,19].

A systematic, direct, and comparative analysis of sand mold and permanent mold casting under identical thermal and compositional conditions is therefore essential to decouple process-governed effects from compositional variables [15,17]. Such a direct comparison provides not only a clearer understanding of process–structure–property relationships but also practical insight for optimizing casting strategies aimed at energy efficiency, material reliability, and sustainable manufacturing [1,2,34].

Accordingly, the present study aims to bridge this knowledge gap by performing a comparative assessment of the microstructural and mechanical behavior of A360 aluminum alloy produced through sand and permanent mold casting routes under identical chemical composition. The specific objectives are: (i) to characterize the microstructural differences particularly in SDAS, eutectic silicon morphology, and porosity arising from the two casting processes; (ii) to evaluate mechanical properties such as tensile strength, yield

strength, elongation, and hardness; and (iii) to establish a direct correlation between casting route, solidification structure, and resulting mechanical performance. The findings are expected to provide both scientific and practical guidance for process optimization in the design and manufacture of high-performance A360 components, contributing to more sustainable and resource-efficient aluminum casting technologies.

2. Materials & Methods

2.1. Alloy Preparation and Casting Procedure

Commercial A360 aluminum alloy was selected for this investigation due to its widespread industrial use and balanced Si–Mg composition, which ensures a favorable combination of castability, mechanical strength, and corrosion resistance [35–37]. The alloy, supplied by Eti Aluminum A.Ş. (Türkiye) under the trade name ETİAL-171, corresponds to AlSi10Mg according to TSE standards, G-AlSi10Mg according to DIN, and A360 in the ASTM classification [38–40].

A360 and its equivalents are extensively used in automotive and aerospace applications owing to their high strength-to-weight ratio, excellent thermal conductivity, and superior fluidity during casting [41–44]. The nominal chemical composition of the alloy used in this study is presented in Table 1.

Table 1. Chemical composition (wt.%) of the A360 aluminum alloy used in this study

Element	wt.%
Al	balance
Si	9.466
Fe	0.404
Cu	0.074
Mn	0.43
Mg	0.438
Zn	0.025

The melting operations were carried out in a Daihan Wisetherm electric resistance furnace (3.6 kW capacity, maximum 1200 °C) equipped with ceramic fiber insulation to ensure thermal stability. High-purity A360 alloy ingots were charged into Mars-Isomelt (A3 type) silicon-carbide crucibles (height 130 mm, base diameter 70 mm, top diameter 110 mm). These crucibles were selected for their rapid heating efficiency, uniform thermal conductivity, and low slag formation [8,45,46]. The alloy was melted at 730 ± 5 °C under an argon-protected atmosphere to minimize oxidation and gas absorption [45-47].

Two distinct casting routes were then applied sand mold casting and permanent mold casting using the same molten batch to eliminate compositional variation. For sand casting, molds were prepared using a silica–bentonite sand mixture with controlled permeability and thermal conductivity, representative of typical industrial sand mold systems. For permanent mold casting, a preheated steel mold (200 °C) was used to promote directional solidification and prevent premature chilling [46,48].

The molten alloy was poured into both molds under identical conditions following gravity-casting principles. After solidification, the castings were allowed to cool in ambient air until room temperature. The obtained ingots were then sectioned to the required dimensions for microstructural characterization and mechanical testing, ensuring identical sampling conditions for both casting routes. The overall melting and solidification protocol followed standard practice for Al–Si–Mg systems, where solidification rate strongly governs microstructure and mechanical performance [48,49].

2.2. Sample Preparation and Metallographic Analysis

Following the solidification process, the gating and riser systems of both sand and permanent mold castings were removed using a band saw to obtain representative specimens. Subsequently, a second cutting operation was performed to prepare samples of appropriate dimensions for microstructural and mechanical characterization. This operation was carried out using a low-speed precision cutting machine equipped with a continuous coolant circulation system. The use of coolant minimized heat accumulation, reduced surface deformation, and allowed the rapid removal of detached particles from the cutting zone, ensuring a smooth and damage-free surface [50,51].

For metallographic preparation, the samples were cold-mounted at room temperature using an epoxy resin–hardener system without applying external pressure. The resin mixture was poured over each specimen and allowed to cure for approximately 2 hours until full polymerization was achieved. After curing, the specimens were removed from the molds and subjected to sequential grinding and polishing processes to obtain flat and reflective surfaces suitable for microscopic examination.

Grinding operations were performed on a Metaserv 2000 grinding–polishing machine using SiC abrasive papers of 200, 400, 600, 800, 1000, and 1200 mesh under running water. Between each step, the specimens were rotated by 90° to remove scratches from the previous stage. Continuous water flow prevented surface overheating and assisted in removing loose abrasive particles. Final polishing was carried out on a velvet cloth using 0.05 μm Al_2O_3 suspension (Buehler, USA) until a mirror finish was achieved. During polishing, the specimens were periodically rotated opposite to the disk rotation direction to improve surface smoothness and reflectivity. Microscopic observations were conducted using an Olympus GX41 optical metallurgical microscope equipped with Clemex Vision image-analysis software.

2.3. Mechanical Testing

Mechanical characterization of the A360 alloy included Brinell hardness**,** tensile**,** and impact tests to evaluate its mechanical performance under static and dynamic loading conditions.

Hardness, although not an intrinsic material property, provides valuable insight into the deformation resistance and quality consistency of cast components. In this study, Brinell hardness testing was conducted using a Zwick/Roell ZHU 187.5 LKV

hardness tester in accordance with ASTM E10**.** A 10 mm tungsten carbide ball indenter and a 4.9 kN load were applied for 15 seconds on both sides of each specimen to ensure measurement uniformity. Prior to testing, the surfaces were carefully ground to eliminate irregularities and ensure a smooth contact interface. The reported hardness value for each condition represents the average of five independent measurements taken at different surface points [50,51].

Tensile testing was performed to determine the yield strength, ultimate tensile strength, and ductility of the alloy. Cylindrical specimens were machined directly from the cast bars and tested in accordance with ASTM E8/E8M standards, which require a gauge length equal to four times the specimen diameter (50 mm in this case) [45,52]. The tests were conducted using a Zwick universal testing machine at room temperature and a crosshead speed of 1 mm/min until fracture occurred. The test setup consisted of upper and lower grips to hold the specimen, a motorized crosshead to apply the tensile load, and a digital data acquisition system to record load–elongation behavior. All tests were repeated three times to ensure reproducibility, and the average results were reported.

Impact behavior of the A360 alloy was evaluated using a computer-controlled JBW-300 Charpy impact tester at room temperature. The tests were performed in accordance with ASTM E23, which defines standardized procedures for determining the impact energy of metallic materials [50].

Rectangular specimens with dimensions of 55 × 10 × 10 mm were prepared, each containing a V-shaped notch with a 45° angle and 2 mm depth machined at the center. During testing, each specimen was simply supported at both ends, while the pendulum hammer was directed to strike the surface opposite the notch root.

The absorbed energy during impact was calculated as the difference between the pendulum's initial potential energy before release and its residual potential energy after fracturing the specimen [50]. This difference represents the energy dissipated by the material through plastic deformation and crack propagation. All measurements were conducted in triplicate, and the mean value was reported for each casting condition to ensure reproducibility and minimize experimental uncertainty.

The Charpy impact results provide a quantitative comparison of toughness and energy absorption capacity between the sand cast and permanent mold cast A360 alloys, illustrating the influence of microstructural refinement, porosity, and solidification rate on their resistance to sudden fracture.

3. Results & Discussion

3.1. Microstructural Observations

Optical microscopy revealed pronounced differences in the solidification morphology of the A360 alloy produced by sand and permanent mold casting.

Figure 1 shows the microstructure of the permanent mold specimen at 50× magnification. The bright regions correspond to the primary α-Al phase, while the darker regions represent the eutectic constituents, primarily consisting of eutectic silicon embedded within the aluminum matrix. The dendritic primary α-Al phase exhibited an relatively coarse dendritic arms, and the interdendritic regions were occupied by dark, needle-like eutectic Si particles. The elongated or rod-like silicon structures observed are typical of eutectic Si formed during the final stages of solidification, consistent with observations in other hypoeutectic Al–Si systems [7,14,17,19,22].

Coarser, blocky phases occasionally detected within the matrix can be attributed either to coarsened eutectic Si or, in some cases, to primary Si particles formed under slow local solidification conditions. Although the formation of primary silicon is uncommon in hypoeutectic Al–Si alloys, it has been reported that such morphologies may appear under prolonged solidification or local solute enrichment [35,46,47,53].

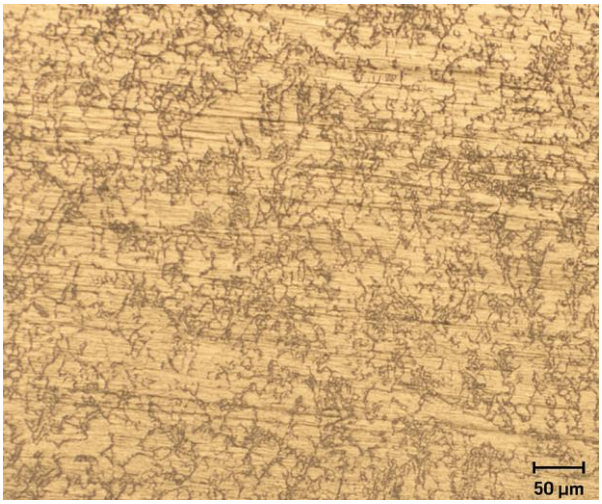


Figure 1. *Optical microstructure of A360 alloy produced by permanent mold casting.*

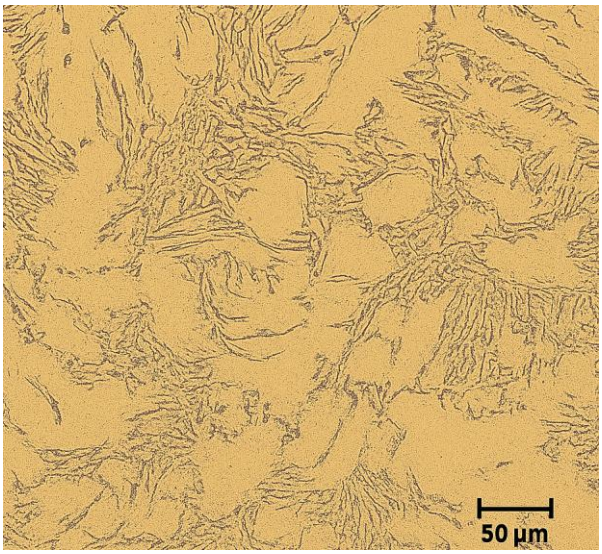


Figure 2. *Optical microstructure of A360 alloy produced by sand mold casting.*

By comparison, Figure 2 illustrates the microstructure of the sand cast A360 alloy, where the slower cooling rate of the sand mold contributed to the growth of coarser dendrites and more interconnected Si networks. The dendritic α-Al phase exhibited wider secondary dendrite arm spacing (SDAS) and larger eutectic Si plates compared to the fine, compact

morphology in the permanent mold specimen. In contrast, the metallic mold solidified more rapidly, producing finer grains and a more homogeneous Si distribution, as also observed by Li et al. [15], Yan et al. [18], and Karakoca and Yılmaz [33].

These microstructural differences clearly demonstrate the effect of cooling rate on phase morphology. Higher solidification rates enhance nucleation and suppress dendritic coarsening, leading to refined α-Al dendrites and spheroidized Si particles. Similar solidification-dependent transitions have been extensively reported in Al–Si–Mg systems [17,19,22,35, 46,47]. The refinement achieved in the permanent mold alloy reduces microsegregation and porosity while improving dislocation compatibility between α-Al and Si, thereby strengthening the alloy and enhancing its overall structural integrity [7,14,19,47,50,54].

Overall, the comparative microstructural evolution presented in Figures 1 and 2 confirms that higher cooling rates achieved in permanent mold casting promote significant grain refinement, suppress eutectic coarsening, and improve morphological uniformity factors directly correlated with the improved mechanical performance of the A360 alloy observed in subsequent sections [15,17,18,19,21,22,46].

3.2. Hardness Results

Following the sectioning of the specimens, both the front and rear surfaces were ground to achieve a smooth finish prior to hardness testing. Measurements were performed on each surface at five different points, and the reported values represent the arithmetic mean of ten independent readings per sample to ensure statistical reliability.

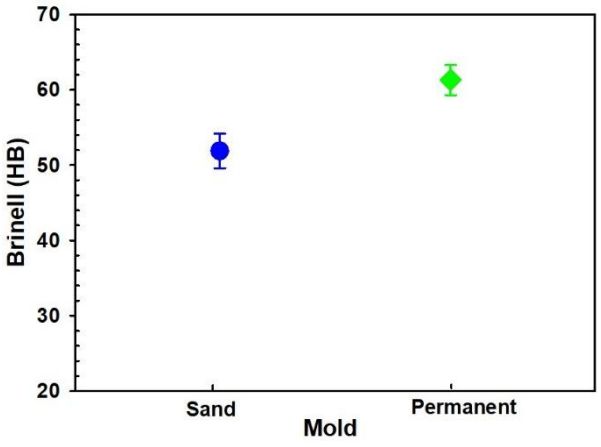


Figure 3. *Comparison of Brinell hardness values of A360 aluminum alloy produced by sand and permanent mold casting.*

Figure 3 presents a comparative overview of the Brinell hardness values for A360 alloys produced by sand and permanent mold casting. According to the results, the permanent mold alloy exhibited an average Brinell hardness of approximately 61.3 HB, while the sand cast alloy showed a lower value of 51.9 HB, corresponding to an improvement of about 18 %. The higher hardness in the permanent mold sample is primarily attributed to the finer eutectic silicon particles and the refined α-Al dendritic network formed under a faster cooling rate. These microstructural features act as effective barriers against dislocation motion, thereby enhancing resistance to localized plastic deformation.

The measured hardness values obtained in this study are consistent with those reported for similar Al–Si alloys, such as A380 and LM6, where enhanced cooling rates and improved eutectic Si refinement have been shown to yield comparable hardness increments [7,10,11,19,21,53,54].

3.3. Tensile Properties

The permanent mold A360 alloy exhibited markedly higher tensile performance compared to its sand cast counterpart. Figure 4 presents the comparative yield strength of A360 aluminum alloys produced via sand and permanent mold casting. The yield strength of the permanent mold specimen was measured as 251.6 MPa, whereas that of the sand cast specimen was 218.7 MPa, confirming the strengthening effect of increased cooling rate during solidification.

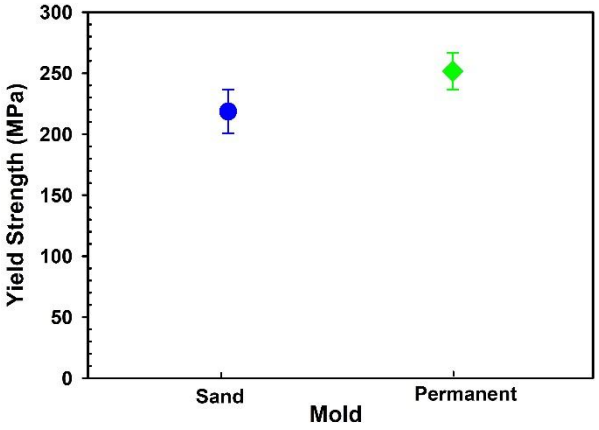


Figure 4. *Comparison of yield strength of A360 aluminum alloy produced by sand and permanent mold casting.*

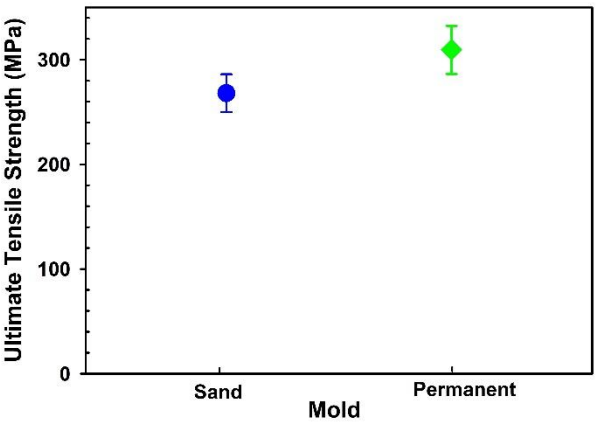


Figure 5. *Comparison of ultimate tensile strength of A360 aluminum alloy produced by sand and permanent mold casting.*

Figure 5 presents a comparative analysis of the ultimate tensile strength values of A360 aluminum alloys produced through sand and permanent mold casting. The ultimate tensile strength (UTS) also increased from 268.1 MPa in the sand cast alloy to 309.4 MPa in the permanent mold alloy.

These findings are consistent with the standard mechanical property ranges reported for A360 alloys and align well with literature data by Li et al. [15], Neuser et al. [17], and Yajjala et al. [19], who similarly observed that enhanced solidification rates promote finer dendritic structures and stronger α-Al/eutectic Si interfaces, thereby improving load-bearing capacity.

The improvement in tensile strength can be attributed to refined secondary dendrite arm spacing (SDAS), improved dendrite–eutectic integrity, and reduced microvoid density, all of which contribute to delaying crack initiation and propagation. The finer microstructure of the permanent mold alloy results in more uniform stress distribution under load, promoting both strength and toughness.

Figure 6 illustrates the comparative elongation (%) of A360 aluminum alloys produced by sand and permanent mold casting. In terms of ductility, the permanent mold specimen exhibited a slightly higher elongation of 3.05 % compared to 2.85 % for the sand cast specimen, corresponding to an approximately 6.6 % improvement. This marginal increase is ascribed to the more homogeneous solidification structure and minimized segregation within the permanent mold. Such microstructural uniformity promotes limited yet more stable plastic deformation prior to fracture, indicating a moderate deviation from the conventional slight ductility retention/improvement discussed by Li et al. [12] and Zhang et al. [22].

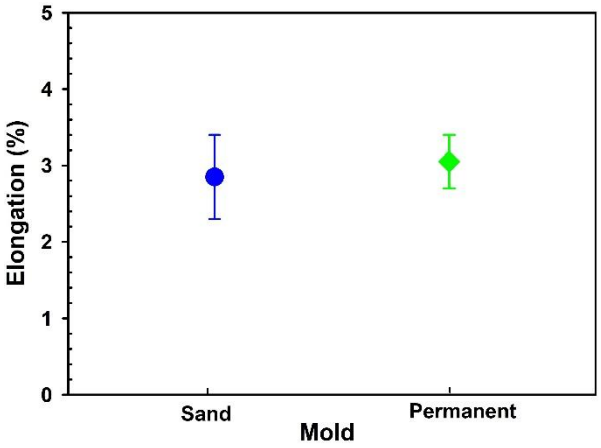


Figure 6. *Comparison of elongation (%) of A360 aluminum alloy produced by sand and permanent mold casting.*

Overall, the mechanical behavior of A360 alloy demonstrates a clear process–structure–property relationship, where higher cooling rates during permanent mold casting lead to refined α-Al dendrites, smaller eutectic Si particles, and enhanced tensile response without significant loss in ductility.

3.4. Impact Energy

The Charpy impact test results revealed that the permanent mold A360 alloy exhibited higher energy absorption capacity (5.07 J) than the sand cast specimen (4.81 J), representing an approximate 5.1 % improvement in impact toughness. Figure 7 presents the comparative Charpy impact energies of A360 aluminum alloys produced via sand and permanent mold casting. This enhancement indicates that the refined microstructure in the permanent mold casting provides more effective energy dissipation during crack propagation, consistent with the strengthening mechanisms observed in other Al–Si–Mg systems [13,17].

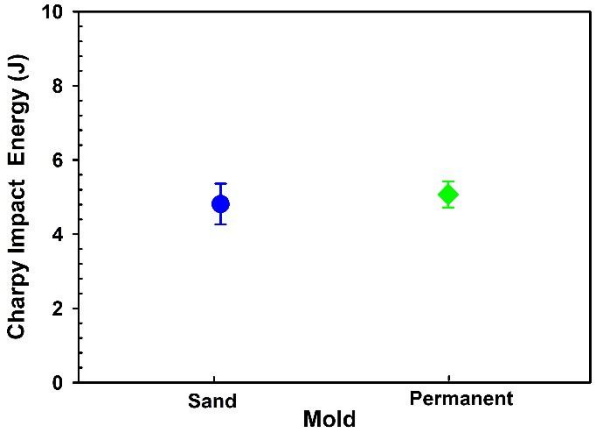


Figure 7. *Comparison of Charpy impact energies of A360 aluminum alloy produced by sand and permanent mold casting.*

The improved toughness of the permanent mold alloy can be attributed to its finer dendritic network and more homogeneous eutectic Si distribution, which collectively reduce microvoid coalescence and promote more stable crack deflection paths. In contrast, the coarser Si particles and irregular interdendritic regions in the sand cast alloy act as stress concentrators, facilitating premature fracture under dynamic loading.

It is well established that the solidification rate significantly affects impact behavior in Al–Si alloys: higher cooling rates promote microstructural refinement, thereby enhancing both strength and toughness [15,22,36,55]. The impact energy levels obtained in this study align with reported values for A360 alloys produced by gravity and permanent mold casting, though they remain slightly lower than those achieved in high-pressure die-cast A360 systems, where the rapid filling and solidification rates can yield impact energies around 5.4 J [56]. Such comparisons underscore that, even under gravity casting conditions, the permanent mold route achieves a considerable improvement in impact performance over sand casting due to its inherently faster solidification kinetics and superior structural integrity.

3.5. Process-Structure-Property Relationship

The comparative analysis distinctly highlights that the intrinsic cooling rate associated with each casting route plays a decisive role in determining the microstructural evolution and, consequently, the mechanical response of the A360 alloy. The permanent mold casting route characterized by its higher thermal gradient and shorter solidification time promotes the formation of finer α-Al dendrites, a more uniformly distributed eutectic Si phase, and significantly reduced porosity. These microstructural refinements collectively enhance the alloy's hardness, strength, and impact toughness by impeding dislocation motion and suppressing crack initiation sites.

The observed improvements confirm a strong and direct process–structure–property correlation, which is a defining

feature of Al–Si–Mg-based systems [13,15,17,22]. The results demonstrate that even in the absence of compositional modification or heat treatment, careful optimization of the casting route alone can yield substantial mechanical enhancement. This emphasizes the metallurgical sensitivity of A360 alloy to solidification kinetics where controlling cooling conditions can achieve an optimal balance between strength, ductility, and toughness thereby underscoring the pivotal role of process design in advancing alloy performance for structural applications.

4. Conclusion

This study systematically examined the influence of casting route sand and permanent mold on the microstructural evolution and mechanical behavior of A360 aluminum alloy produced under identical chemical composition and processing conditions.

a) Microstructure: The permanent mold alloy exhibited a refined α-Al dendritic morphology with a significantly smaller secondary dendrite arm spacing (SDAS) and a more homogeneous eutectic Si network compared with the coarse, irregular microstructure of the sand cast alloy. The accelerated cooling in the permanent mold promoted uniform nucleation, restricted dendritic coarsening, and reduced porosity formation.

b) Mechanical properties: The permanent mold specimens demonstrated measurable improvements in mechanical performance approximately 18% higher hardness and 10–15% higher tensile strength relative to the sand cast alloy. These enhancements are primarily attributed to microstructural refinement, improved α-Al/eutectic Si interface bonding, and suppression of microvoid initiation sites.

c) Impact behavior: The impact energy absorption capacity of the permanent mold alloy was about 5% higher than that of the sand cast specimen, confirming superior toughness and resistance to crack propagation. The enhanced structural uniformity facilitated more efficient energy dissipation during dynamic fracture.

d) Overall insight: The findings conclusively establish that the solidification rate inherent to the casting process serves as a critical metallurgical control parameter governing the process–structure–property relationship in A360 alloys. Even in the absence of compositional or thermal modifications, optimization of casting route alone can yield substantial mechanical benefits.

This work provides a robust experimental framework and baseline for future studies focusing on microalloying, modification treatments, and sustainable optimization of A360 alloy systems, particularly in the context of high-integrity light-metal casting for automotive and aerospace applications. Future research may build upon the present findings by investigating targeted microalloying strategies to modulate phase stability, implementing advanced post-solidification heat-treatment pathways to optimize precipitate evolution and mechanical response, and undertaking systematic comparisons with high-pressure die-casting to more rigorously elucidate the underlying process–structure–property relationships in A360 alloy systems.

Author Contributions

Research – Eda ERGÜN SONGÜL (EES), Derya DIŞPINAR (DD); Experimental performance – EES, DD; Data collection – EES; Data processing – EES; Literature review – EES; Writing – EES, DD; Review and editing – EES, DD.

Conflict of Interest Statement

The authors declare that there is no conflict of interest with respect to the research, authorship, and/or publication of this article.

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